

DTSA 5511

Introduction to Deep Learning

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	AI Engineer (44.5 / 100)	36.7 / 100
SPECIALIZATION Machine Learning: Theory and Hands-on Practice with Python		

37

CURRENT AVG SCORE
across 6 job roles

79

POTENTIAL SCORE
with recommended additions

+51

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Course will cover the basics of deep learning, such as multilayer perceptron, convolutional neural network, recurrent neural network, how to build and train neural network models, optimization methods, and application examples. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	44.5	89.1	+44.6	HIGH
Machine Learning Engineer	42.0	78.8	+36.7	HIGH
Applied Scientist	38.6	83.6	+45.0	HIGH
Data Scientist	37.8	89.1	+51.3	HIGH
ML Engineer	33.2	70.4	+37.2	HIGH
MLOps Engineer	23.9	65.7	+41.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Deep Learning 2 of 6 roles

cnn

training

Business Communication 1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 MEDIUM Consider Deep Learning content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.

3 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

AI Engineer

44.5 → 89.1 (+44.6 pts)

data

engineering

form

software

systems

problems

technology

detection

Machine Learning Engineer

42.0 → 78.8 (+36.7 pts)

data

engineering

tools

exploratory

time

design

science

applications

Applied Scientist

38.6 → 83.6 (+45.0 pts)

lab

research

literature

problems

data

systems

industry

engineering

Data Scientist

37.8 → 89.1 (+51.3 pts)

data

product

evolution

insights

different

social

engineering

problems

ML Engineer

33.2 → 70.4 (+37.2 pts)

cnn

data

systems

engineering

starting

problems

impact

form

MLOps Engineer

23.9 → 65.7 (+41.8 pts)

detection

data

text

systems

image

engineering

training

tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.